SHAPE
Smart City Higher Education Advancement Programme
IN EAST AFRICA

PROJECT BOOK · OCTOBER 2026



ERASMUS+ CAPACITY BUILDING IN HIGHER EDUCATION

Shaping Smarter Cities through Higher Education

The Smart City Higher Education Advancement Program in East Africa — nine universities, three countries of impact, one shared mission.

Kenya · Somalia · Uganda · Estonia · Germany · Lithuania

Co-funded by the Erasmus+ Programme of the European Union

2025 – 2028 · ERASMUS-EDU-2025-CBHE-STRAND-2

INSIDE

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This book tells the SHAPE story in eleven short chapters — why smart city education matters for East Africa, what the consortium is building, and how we will know it worked.

> *"Programmes that reflect the realities and priorities of African cities rather than replicating models developed elsewhere."*

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CHAPTER 01

SHAPE at a Glance

An Erasmus+ Capacity Building initiative connecting European and East African universities to advance smart, sustainable city development.

36 months

Project Duration

Three years, from curriculum design and staff training through to a self-sustaining regional alliance.

ERASMUS-EDU-2025-CBHE-STRAND-2

9 partners

Partner Institutions

Nine universities, coordinated by Moi University, pairing European technical expertise with East African context.

Coordinator: Moi University

3 countries

East African Focus

Higher education institutions and city stakeholders in Kenya, Somalia, and Uganda are the primary beneficiaries.

Kenya · Somalia · Uganda

12 areas

Thematic Focus

From AI and open data to smart mobility, energy, health, and community engagement in city planning.

12 smart city domains

Erasmus+

EACEA Funded

Backed by the European Education and Culture Executive Agency under Capacity Building in Higher Education.

Strand 2 · CBHE

1 alliance

Regional Alliance

A new EU–East Africa Smart Cities Alliance keeps universities, industry, and government collaborating long-term.

Lasting beyond 2028

CHAPTER 02

The Challenge

The Smart City Higher Education Advancement Program in East Africa (SHAPE) is an Erasmus+ Capacity Building in Higher Education initiative designed to address the growing shortage of smart city skills across East Africa, particularly in Kenya, Somalia, and Uganda. Rapid urbanization in the region has created increasing demand for professionals capable of planning, designing, and managing smart, sustainable, and inclusive cities. However, higher education institutions currently lack specialized curricula, qualified academic staff, practical training facilities, and strong collaboration with industry and public institutions.

SHAPE responds to these challenges by modernizing higher education through innovative curricula, academic capacity building, digital learning, research collaboration, and open innovation. The project combines the technical expertise of European universities with the contextual knowledge of East African institutions to develop educational programmes that reflect the realities and priorities of African cities rather than replicating models developed elsewhere.

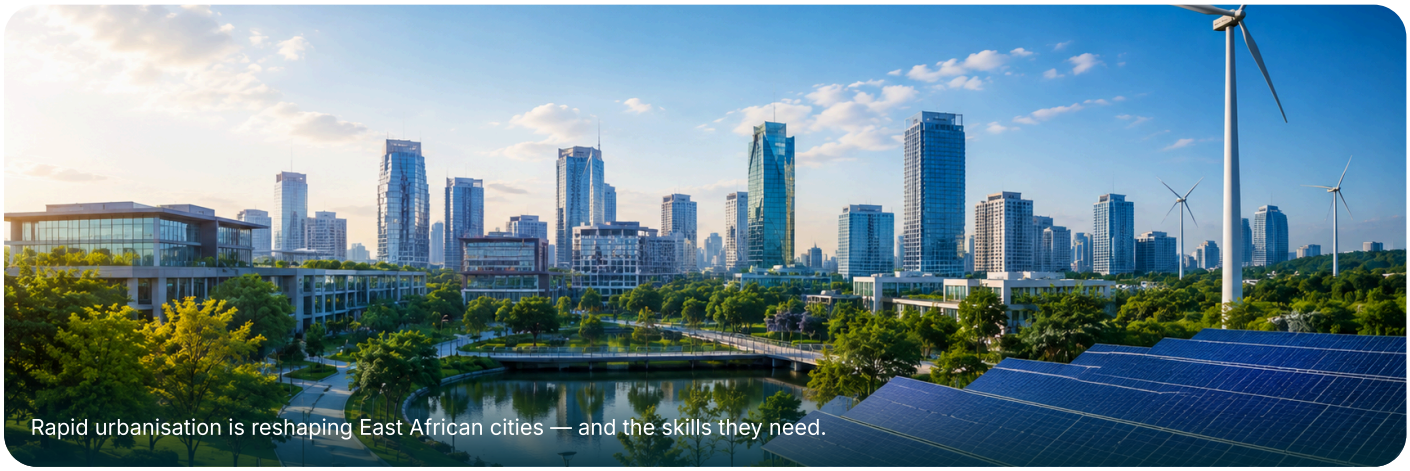
93%

of local governments in Africa report skills shortages as a major barrier to smart city development.

29%

of African cities currently have a smart city strategy in place.

Source: UN-Habitat, as cited at the SHAPE launch.



Rapid urbanisation is reshaping East African cities — and the skills they need.

CHAPTER 03

Our Response

The project will introduce new Master's level courses and micro-credentials covering key smart city domains, including Artificial Intelligence, Open Data, Smart Mobility, Smart Energy, Smart Health, Community Engagement, and Sustainable Urban Development. Academic staff will receive specialized training, while universities will establish Open Innovation Labs to promote entrepreneurship, applied research, hackathons, and collaboration with industry and government.

Beyond curriculum development, SHAPE aims to create lasting partnerships between universities, municipalities, industry, civil society, and policymakers through a new EU–East Africa Smart Cities Alliance. By strengthening higher education institutions and producing highly skilled graduates, the project contributes to sustainable economic growth, digital transformation, and resilient urban development throughout East Africa. The initiative also aligns closely with the EU Global Gateway Strategy, supporting digital innovation, green transition, sustainable transport, and inclusive governance across Sub-Saharan Africa.

Our Goal

To strengthen higher education institutions in East Africa by developing innovative academic programmes, practical innovation ecosystems, and skilled graduates capable of designing, implementing, and sustaining smart, inclusive, and sustainable cities across Kenya, Somalia, and Uganda.

Our Mission

To modernize higher education through innovative curricula, digital technologies, industry collaboration, and capacity building that empower universities to produce professionals with the knowledge, skills, and leadership needed to drive sustainable smart city development in East Africa.

New curricula
Master's courses & micro-credentials

Stronger staff
Train-the-trainer programmes

Innovation labs
Hackathons & applied research

Lasting alliance
EU–East Africa Smart Cities Alliance

CHAPTER 04

Twelve Thematic Areas

SHAPE's curricula and Open Innovation Labs span twelve interconnected domains shaping the future of smart, sustainable cities.

01
Smart Cities & Sustainable Urban Development

02
Artificial Intelligence & Data-Driven Decision Making

03
Open Data & Digital Governance

04
Smart Energy & Renewable Energy Systems

05
Smart Mobility & Intelligent Transport Systems

06
Smart Health & Digital Healthcare

07
Internet of Things (IoT) & Sensor Technologies

08
Community Engagement & Citizen Participation

09
Innovation & Entrepreneurship

10
Digital Transformation in Higher Education

11
Open Innovation Labs & Hackathons

12
Capacity Building & Curriculum Development

CHAPTER 05

Who We Serve

SHAPE's impact reaches every part of the smart city ecosystem — from classrooms to city halls.

- Universities & HEIs in Kenya, Somalia & Uganda
- Academic Staff, Researchers & Administrators
- Undergraduate & Postgraduate Students
- Government Institutions & Municipalities
- Smart City Practitioners & Urban Planners
- Technology Companies & Innovation Hubs
- Entrepreneurs & Startups
- Civil Society Organizations
- Local Communities in Urban Development
- Public & Private Sector Stakeholders



Partners and stakeholders at the Mombasa partners meeting, 2026.

CHAPTER 06

The Consortium

COORDINATOR

Moi University, Kenya

Prof. Kirimi Kiriamiti · Project Coordinator

University of Tartu

Estonia

Otto-von-Guericke-University Magdeburg

Germany

Red Sea University

Somalia

Open University of Kenya

Kenya

Mogadishu University

Somalia

Makerere University

Uganda

Kampala International University

Uganda

Lithuanian University of Health Sciences

Lithuania

Associated partners

- Konza Technopolis Development Authority · Kenya
- Université du Burundi · Burundi
- Institute of Social Policy and Research · South Sudan

CHAPTER 07

Work Plan

Five coordinated Work Packages carry SHAPE from research through to a lasting EU–East Africa alliance.

WORK PACKAGE 1 · M1–M36

Project Management & Coordination Activities

Oversees day-to-day coordination, quality assurance, and reporting to keep SHAPE on schedule and aligned across every partner institution.

Lead: **Moi University**

01

WORK PACKAGE 2 · M2–M12

Establishing Design Requirements for Smart Cities in Sub-Saharan Africa

Assesses regional needs and educational gaps to shape a strategic roadmap for a smart city curriculum built for East Africa's realities.

Lead: **Open University of Kenya**

02

WORK PACKAGE 3 · M10–M33

Building a New Educational Curriculum for Smart Cities

Develops and pilots new Master's-level courses, open online learning, and trainer programmes spanning smart city disciplines.

Lead: **Kampala International University**

03

WORK PACKAGE 4 · M25–M34

Implementation of Smart City Acceleration & Open Innovation Programme

Establishes Open Innovation Labs and hackathons, and launches the EU–East Africa Smart City Alliance, turning research into practice.

Lead: **University of Tartu**

04

WORK PACKAGE 5 · M1–M36

Dissemination, Exploitation, Sustainability & Communication

Builds SHAPE's public presence and long-term sustainability through communication strategy, outreach, and digital channels.

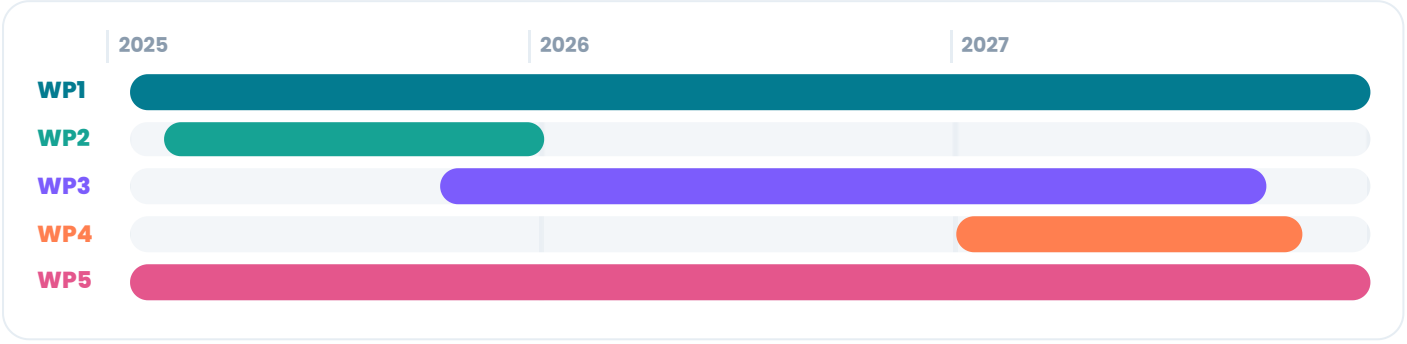
Lead: **Mogadishu University**

05

CHAPTER 08

Timeline & Deliverables

36 months, five Work Packages and 20 deliverables.



WP1

D1.1
Data management plan

D1.2
Progress Report to EU

D1.3
Quality assurance: MEAL — monitoring, evaluation, accountability and learning strategy

WP2

D2.1
Report on needs and competencies for smart cities in East Africa

D2.2
Report on educational reform needs in partner universities in East Africa

D2.3
Strategic roadmap for East African partner universities to develop a comprehensive smart city curriculum

WP3

D3.1
Master's-level new and updated educational curriculum on smart cities and their different aspects (e.g. smart energy, smart health etc.)

D3.2
Open online courses

D3.3
Train the Trainer sandbox — first cohort of trained trainers

D3.4
First cohort of students from pilot courses — feedback and refinement

D3.5
Accreditation of developed micro-credentials and updated master's programmes

WP4

D4.1
Handbook on training of trainers for open innovation / smart city labs

D4.2
Open innovation labs in partner universities of East Africa

D4.3
3 Open Innovation Hackathons

D4.4
Initiation document for the EU–East Africa Smart City Alliance

D4.5
Report on virtual exchanges

WP5

D5.1
Dissemination, communication and networking strategy

D5.2
Website and social media pages development

D5.3
Sustainability strategy

D5.4
List of outreach activities

CHAPTER 09

Expected Impact

What success looks like when SHAPE reaches the end of its three-year journey — for institutions, graduates, and the region.

01 New Accredited Curricula

New accredited smart city curricula and micro-credential programmes introduced across partner universities.

02 Stronger Academic Staff

Academic staff equipped with modern teaching methodologies and advanced smart city competencies.

03 Open Innovation Labs

Open Innovation Labs established within East African universities to promote innovation, entrepreneurship, and applied research.

04 Deeper Collaboration

Stronger collaboration between universities, governments, industry, and local communities in addressing urban development challenges.

05 Employable Graduates

Enhanced digital, technical, and entrepreneurial skills among students and graduates, improving employability and regional economic growth.

06 International Collaboration

Increased international collaboration through virtual exchanges, joint research, and knowledge sharing between European and East African institutions.

07 A Lasting Alliance

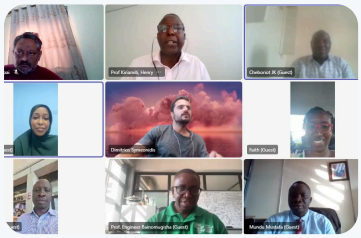
A sustainable EU–East Africa Smart Cities Alliance established to continue collaboration beyond the project's lifetime.

08 Long-Term Regional Impact

Long-term contribution to inclusive, resilient, and sustainable smart city development across East Africa through higher education transformation.

CHAPTER 10

The Journey So Far



18 JUNE 2026 · ONLINE

SHAPE Project Official Launch

Consortium partners from Kenya, Uganda, Somalia, Estonia, Germany and Lithuania met to establish a shared vision and align on the implementation plan.



28 SEPTEMBER 2026 – 2 OCTOBER 2026 · MOMBASA, KENYA

International Partners Meeting

Hosted by Moi University: partners aligned on objectives, launched key work packages, visited Fort Jesus and the Mama Ngina Waterfront, and held technical workshops.













→ Next: Train-the-Trainer sandbox, three Open Innovation Hackathons and the first pilot-course cohort (2027).

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IN EAST AFRICA

Get in Touch

Join us in shaping smarter, more inclusive cities across East Africa.

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Hosted by OPEN UNIVERSITY OF KENYA

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